

Major Project Report: Enterprise AI Knowledge Hub

Retrieval-Augmented Generation (RAG) System using LangChain & Gemini

1. Executive Summary

The Enterprise AI Knowledge Hub is an intelligent document assistant designed to help organizations search and extract insights from their data. Utilizing a Retrieval-Augmented Generation (RAG) architecture, the system parses multi-format documents (PDF, DOCX, TXT, MD, HTML), indexes their contents into a high-performance local vector database (ChromaDB), and generates context-aware responses using the Google Gemini model. All answers are accompanied by verifiable source page citations to ensure truthfulness and eliminate LLM hallucinations.

2. Key System Features

- **Multi-Format Ingestion:** Supports Word documents (.docx), PDFs (.pdf), plain text (.txt), Markdown (.md), and HTML files.
- **Persistent ChromaDB Store:** Local database vector search using semantic embeddings generated by `models/gemini-embedding-001`.
- **Conversational Retrieval:** Remembers previous questions in the session to handle follow-up queries naturally.
- **Source Citation Expansion:** Shows the document name, page number, and text snippet for every generated response.
- **Dynamic Document Manager:** Allows users to upload and delete documents dynamically on the sidebar interface.

3. Technical Architecture

The system pipeline consists of the following components:

1. **Document Loader:** Routes documents by file extension to appropriate LangChain parsers.
2. **Text Splitter:** Chunks data using `RecursiveCharacterTextSplitter` (size: 1000, overlap: 200).
3. **Vector Indexing:** Saves document embeddings in a persistent directory `./chroma_db`.
4. **Conversational Chain:** Uses `gemini-3.5-flash` to synthesize answers using chat history and retrieved context.

4. User Interface Dashboard

The web application features a sleek, premium dark-mode dashboard created with Streamlit:

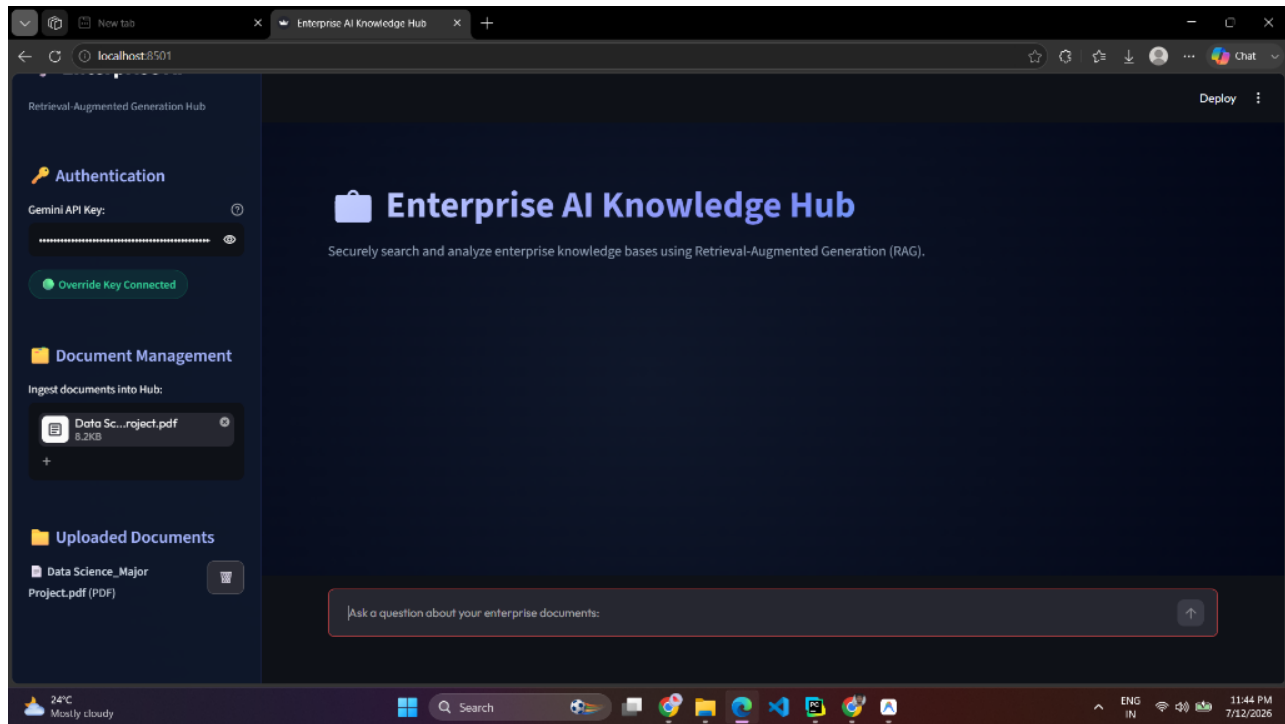


Figure 4.1: Deployed Streamlit Dashboard Interface

5. Setup and Execution Instructions

To run the application locally, use the following commands:

```
git clone https://github.com/Amalbinu123/Enterprise_AI_Knowledge_Hub.git
cd Enterprise_AI_Knowledge_Hub
pip install -r requirements.txt
streamlit run app.py
```

Evaluation Note: The project can be run using the live URL:

<https://enterpriseaiknowledgeapp-fef3tkgx4qpguycyyhz2e8.streamlit.app/>. Evaluators can paste their own Gemini API key directly into the sidebar to test it instantly.